HOSPITAL EMERGENCY RESPONSE CO-ORDINATOR

SOURCE CODE

Triage Process

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#include "hospital_messages.h"


int main(void) {
    setvbuf(stdout, NULL, _IONBF, 0);
    name_attach_t *attach = name_attach(NULL, ATTACH_POINT_TRIAGE, 0);
    if (attach == NULL) {
        perror("[TriageProcess] name_attach failed");
        return EXIT_FAILURE;
    }

    printf("[TriageProcess] Online and listening on '%s' (chid=%d)\n",
           ATTACH_POINT_TRIAGE, attach->chid);
    fflush(stdout);

    int priority_coid = -1;
    hospital_msg_t msg;
    hospital_reply_t rep;
    int rcvid;

    while (1) {
        rcvid = MsgReceive(attach->chid, &msg, sizeof(msg), NULL);
        if (rcvid == -1) {
            perror("[TriageProcess] MsgReceive error");
            continue;
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    }
    if (rcvid == 0) {
        continue;
    }

    if (msg.msg_type == MSG_TYPE_BATCH_DONE) {
        printf("[TriageProcess] Batch complete signal received. Forwarding to
PriorityProcess...\n");
        fflush(stdout);

        if (priority_coid == -1) {
            priority_coid = name_open(ATTACH_POINT_PRIORITY, 0);
        }
        if (priority_coid != -1) {
            hospital_reply_t fwd_rep;
            MsgSend(priority_coid, &msg, sizeof(msg), &fwd_rep, sizeof(fwd_rep));
        }

        memset(&rep, 0, sizeof(rep));
        rep.msg_type = MSG_TYPE_REPLY_OK;
        rep.status = 0;
        strncpy(rep.message, "Triage batch forwarded", sizeof(rep.message) - 1);
        MsgReply(rcvid, EOK, &rep, sizeof(rep));
        continue;
    }
    printf("[TriageProcess] Received Patient %s (%s, Severity: %d)\n",
           msg.record.patient.patient_id,
           msg.record.patient.condition,
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            msg.record.patient.severity);
        fflush(stdout);
        if (priority_coid == -1) {
            priority_coid = name_open(ATTACH_POINT_PRIORITY, 0);
            if (priority_coid == -1) {
                printf("[TriageProcess] Warning: Could not connect to '%s' yet (errno=%d).
Retrying...\n",
                        ATTACH_POINT_PRIORITY, errno);
                fflush(stdout);
            }
        }
        if (priority_coid != -1) {
            msg.msg_type = MSG_TYPE_TRIAGE_FORWARD;
            hospital_reply_t prio_rep;
            if (MsgSend(priority_coid, &msg, sizeof(msg), &prio_rep, sizeof(prio_rep)) == -1) {
                perror("[TriageProcess] MsgSend to PriorityProcess failed");
                name_close(priority_coid);
                priority_coid = -1;
            }
        }
        memset(&rep, 0, sizeof(rep));
        rep.msg_type = MSG_TYPE_REPLY_OK;
        rep.status = 0;
        strncpy(rep.message, "Admitted & Forwarded to Priority", sizeof(rep.message) - 1);
        MsgReply(rcvid, EOK, &rep, sizeof(rep));
    }
    name_detach(attach, 0);
    return EXIT_SUCCESS;
```